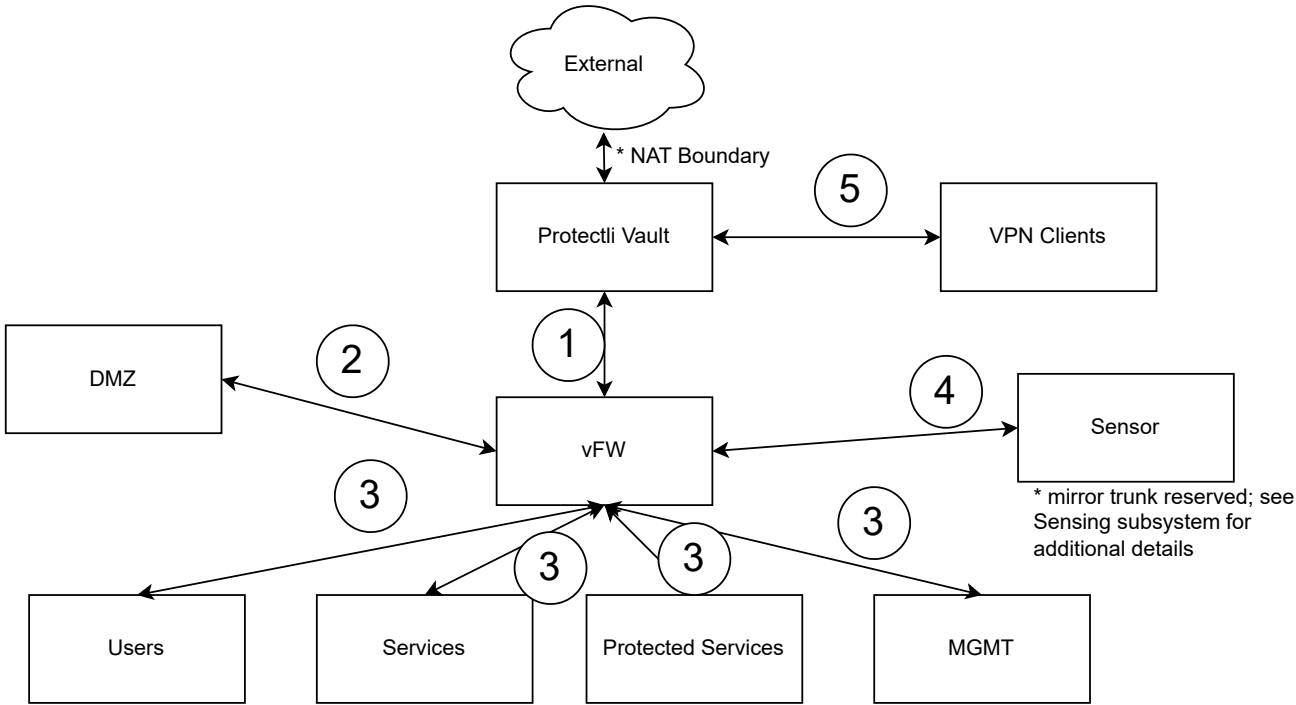


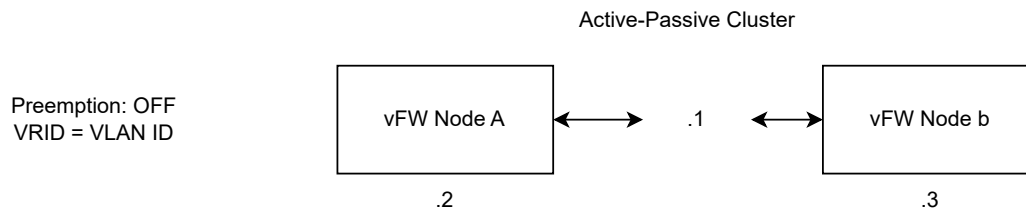
Network Stack Subsystem			
Subsystem Code	ns	Component Codes	core = Core Switch ewfw = East-West/Inner Firewall nsfw = North-South/Outer Firewall



Data Flows			
ID	Port/Proto	Description	Controls
1	All	DMZ Transit traffic between the E-W and N-S policy evaluation points. Static routes ensure connectivity for internal networks	Small subnet (/29) and no servers limits opportunity for exploitation. No management interfaces available.
2	All	Traffic destined for the DMZ.	See DMZ subsystems for specific controls.
3	All	Internal traffic, either from the DMZ after being evaluated or stateful return traffic.	See subsystems for specific controls.
4	All	All SDN segments mirror traffic to the sensor on a dedicated promiscuous interface.	Sensor interface will run only in promiscuous mode; will not have an IP that can be targetted for exploitation and will receive mirrored packets only.
5	All	VPN Clients enter from external via WireGuard running on Protectli Vault UDP/51820, are decrypted, and then re-routed using pre-defined client source IPs. Split-tunneling only processes traffic destined for inside the domain.	Public Key authentication - clients receive the pfSense public key and pfSense receives the client public key for mutual authentication. vFWs enforce restriction to only externally-eligible subnets (VLAN 20)

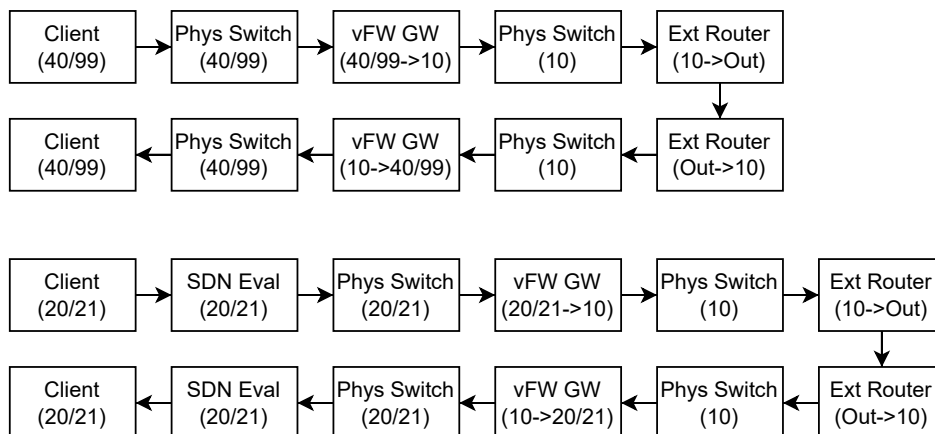
* Each L2 domain has additional SDN restrictions in place per subsystem. Allowed flows are declarative, and detailed in each subsystem architectural diagrams.

vFW Architecture

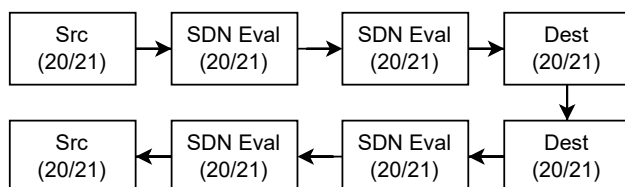


L2 Paths

- Internal to External

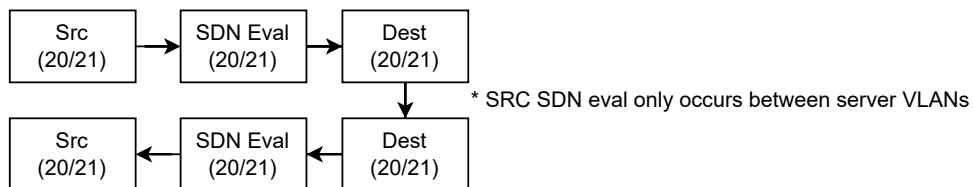


- Internal to Internal (Intra-VLAN)



- Internal to Internal (Inter-VLAN)

* Phys Switch may carry trunked traffic depending on src/dest location in the ProxMox cluster



- External to Internal

